Deep Learning for Automatic Modulation Classification: A Comparative Study of CNN, LSTM, and Hybrid Architectures

Nabeegh Khan

March 2026

Abstract

This report presents a comparative study of deep learning architectures for automatic modulation classification (AMC) using the RadioML 2016.10a dataset. Three architectures were implemented and evaluated: a 1D Convolutional Neural Network (CNN), a Long Short-Term Memory (LSTM) network, and a hybrid CNN-LSTM model. The CNN achieved the highest overall accuracy of 49.8%, reaching 72.7% at high SNR conditions (≥0 dB). In contrast, the pure LSTM failed to learn meaningful features from raw I/Q data, achieving only 11.1% accuracy (near random chance). The hybrid model achieved 21.1% overall accuracy, demonstrating that while CNN feature extraction enables learning, the addition of LSTM layers does not improve performance for this task. These results suggest that spatial feature extraction via convolution is more effective than temporal modeling for raw I/Q-based modulation classification.

1. Introduction

Automatic modulation classification (AMC) is a critical component in cognitive radio systems, spectrum monitoring, and intelligent wireless communication networks. The ability to automatically identify the modulation scheme of a received signal enables dynamic spectrum access, interference management, and adaptive communication protocols. Traditional AMC approaches rely on expert-designed features such as higher-order statistics, cyclostationary features, or wavelet transforms. However, these methods require significant domain expertise and may not generalize well across different channel conditions.

Deep learning offers a promising alternative by automatically learning discriminative features directly from raw signal data. Recent advances have demonstrated the effectiveness of convolutional neural networks (CNNs) for processing raw in-phase and quadrature (I/Q) samples. This project investigates three deep learning architectures for AMC: a 1D-CNN that treats the I/Q signal as a multi-channel 1D sequence, an LSTM network designed to capture temporal dependencies, and a hybrid architecture that combines CNN feature extraction with LSTM sequence modeling.

The primary research questions addressed are: (1) How effective are different neural network architectures for raw I/Q-based modulation classification? (2) Does incorporating temporal modeling via LSTM improve classification performance over pure convolutional approaches? (3) What SNR threshold is required for reliable classification?

2. Preliminaries and Problem Formulation

The AMC problem can be formulated as a multi-class classification task. Given a received signal $r(t) = s(t) * h(t) + n(t)$, where $s(t)$ is the transmitted signal, $h(t)$ is the channel response, and $n(t)$ is additive noise, the goal is to identify the modulation scheme M from a set of K possible modulation types $\{M_1, M_2, ..., M_K\}$.

In this study, the signal is represented as discrete I/Q samples: $x[n] = [I[n], Q[n]]$ for $n = 1, 2, ..., N$, where $I[n]$ and $Q[n]$ are the in-phase and quadrature components respectively. Each sample consists of $N = 128$ time steps with 2 channels (I and Q), resulting in an input tensor of shape (2, 128). The classification objective is to learn a mapping $f: X \to Y$ where $X \in R^{2\times128}$ and $Y \in \{0, 1, ..., 10\}$ represents one of 11 modulation classes.

2.1 Dataset

The RadioML 2016.10a dataset is a synthetic dataset generated using GNU Radio, containing 220,000 signal samples across 11 modulation types: 8PSK, AM-DSB, AM-SSB, BPSK, CPFSK, GFSK, PAM4, QAM16, QAM64, QPSK, and WBFM. Each modulation type includes samples at 20 different signal-to-noise ratio (SNR) levels ranging from -20 dB to +18 dB in 2 dB increments. The dataset was split into training (72%), validation (8%), and test (20%) sets using stratified sampling to maintain class and SNR balance.

3. Design

Three neural network architectures were implemented to investigate different approaches to feature extraction and classification from raw I/Q data.

3.1 1D Convolutional Neural Network (CNN)

The CNN architecture processes the 2-channel I/Q input through two convolutional layers, each with 64 filters and kernel size 8. Max pooling with stride 2 follows each convolution, reducing the temporal dimension from 128 to 32. The flattened features pass through a fully connected layer with 128 units before the final 11-class softmax output. Dropout (p=0.5) is applied for regularization. ReLU activation is used throughout except for the output layer.

3.2 Long Short-Term Memory (LSTM)

The LSTM architecture treats the I/Q samples as a sequence of 128 time steps with 2 features per step. A single LSTM layer with 64 hidden units processes the sequence, and the final hidden state is passed to a fully connected layer for classification. This design tests whether temporal dependencies in the I/Q sequence provide useful information for modulation classification.

3.3 Hybrid CNN-LSTM

The hybrid architecture combines CNN feature extraction with LSTM sequence modeling. A single convolutional layer (64 filters, kernel size 8) with max pooling extracts local features, reducing the input to a (64, 64) feature map. This is then processed by an LSTM layer (64 hidden units) that models dependencies across the extracted features. The final hidden state feeds into the classification layer.

4. Methodology

4.1 Implementation

All models were implemented in PyTorch. Training used the Adam optimizer with learning rate 0.001 and cross-entropy loss. The CNN was trained for 15 epochs on the full training set (158,400 samples), while the LSTM and hybrid models were trained for 10 epochs on a subset (50,000 samples) due to computational constraints. Batch size was 128 for the CNN and 256 for LSTM-based models.

4.2 Evaluation Metrics

Models were evaluated on the held-out test set (44,000 samples) using overall classification accuracy, per-SNR accuracy curves, and confusion matrices. Per-SNR analysis is particularly important for understanding model performance under different channel conditions, as low SNR environments present significantly greater classification challenges.

5. Numerical Experiments

5.1 Overall Results

Table 1 summarizes the classification accuracy of each model across different SNR regimes. The CNN significantly outperforms both the LSTM and hybrid architectures, achieving 49.8% overall accuracy compared to 11.1% and 21.1% respectively.

Model	Overall Accuracy	High SNR (≥0 dB)	Low SNR (<0 dB)
CNN	49.8%	72.7%	26.6%
LSTM	11.1%	13.0%	9.2%
Hybrid	21.1%	24.8%	17.5%

Table 1: Classification accuracy comparison across SNR regimes

5.2 SNR Analysis

Figure 1 shows the classification accuracy as a function of SNR for all three models. The CNN exhibits a characteristic S-curve, with accuracy rising sharply between -10 dB and 0 dB before plateauing around 74% at high SNR. The LSTM remains essentially flat across all SNR levels, indicating a fundamental inability to learn discriminative features from the raw I/Q representation. The hybrid model shows modest improvement over the LSTM, suggesting that the CNN component provides some useful feature extraction, though the overall performance remains limited.

[Figure 1: Model comparison accuracy vs. SNR - see figures/model_comparison.png in repository]

5.3 Confusion Analysis

Analysis of the CNN confusion matrix at high SNR (≥10 dB) reveals specific classification challenges. Modulations with similar constellation structures show significant confusion: QAM16 achieves only 80% accuracy with 18% misclassified as QAM64, while QAM64 reaches 85% with 13% confused as QAM16. The most problematic case is QPSK, achieving only 29% accuracy at high SNR with 60% misclassified as 8PSK. This reflects the inherent similarity between QPSK (4-point constellation) and 8PSK (8-point constellation) signal representations. Frequency-based modulations (CPFSK, GFSK) achieve near-perfect classification at high SNR.

6. Discussion

The results demonstrate that convolutional architectures are fundamentally better suited for raw I/Q-based modulation classification than recurrent architectures. Several factors explain this finding:

Local Pattern Recognition: CNNs excel at detecting local patterns in the I/Q signal that correspond to modulation-specific characteristics such as constellation geometry and phase transitions. The convolutional filters effectively learn these discriminative features without explicit feature engineering.

LSTM Limitations: The pure LSTM's failure suggests that long-range temporal dependencies are not the primary discriminative factor for modulation classification. The I/Q samples within a 128-sample window may not exhibit the sequential structure that LSTMs are designed to capture. Additionally, LSTMs require larger datasets and longer training to converge effectively.

Hybrid Performance: The hybrid model's intermediate performance confirms that CNN-based feature extraction is beneficial, but adding LSTM layers does not improve—and may actually hinder—classification performance compared to a pure CNN approach.

SNR Threshold: All models show minimal discrimination below -10 dB, suggesting a fundamental limit on classification performance in very low SNR conditions. Practical AMC systems should consider this threshold when deploying deep learning solutions.

7. Conclusions

This study compared three deep learning architectures for automatic modulation classification using the RadioML 2016.10a dataset. Key findings include:

1. The 1D-CNN architecture achieves the best performance (49.8% overall, 72.7% at high SNR), demonstrating the effectiveness of convolutional feature extraction for raw I/Q data.
2. Pure LSTM networks fail to learn meaningful features from raw I/Q samples, achieving only random-chance accuracy.
3. Hybrid CNN-LSTM architectures do not improve upon pure CNN performance for this task.
4. Classification becomes unreliable below -10 dB SNR regardless of architecture.

Future work could explore deeper CNN architectures (e.g., ResNet), attention mechanisms, or alternative signal representations such as spectrograms or constellation diagrams. Data augmentation techniques specific to wireless signals may also improve generalization.

References

[1] O'Shea, T.J., Corgan, J., Clancy, T.C. (2016). "Convolutional Radio Modulation Recognition Networks." In Proceedings of the International Conference on Engineering Applications of Neural Networks (EANN).

[2] O'Shea, T.J., West, N. (2016). "Radio Machine Learning Dataset Generation with GNU Radio." In Proceedings of the GNU Radio Conference.

[3] West, N.E., O'Shea, T.J. (2017). "Deep Architectures for Modulation Recognition." In IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN).

[4] Goodfellow, I., Bengio, Y., Courville, A. (2016). Deep Learning. MIT Press.

[5] Hochreiter, S., Schmidhuber, J. (1997). "Long Short-Term Memory." Neural Computation, 9(8), 1735-1780.

Use of AI

This project was developed with extensive assistance from Claude (Anthropic). The AI provided guidance on project structure, wrote and debugged PyTorch code, explained deep learning concepts, and assisted with documentation including this report. The author directed the project goals, selected the research questions, executed all code on a local machine, and made decisions about which results to include and how to interpret them. This collaboration reflects an emerging workflow in technical projects where AI tools augment human capabilities—the author learned deep learning concepts through the process while leveraging AI to accelerate implementation and documentation.